2/23/22

Monday, February 21, 2022 11:23 AM

M168_Notes_6.2

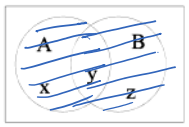
6.2 The Number of Elements in a Finite Set

The number of elements in set A is $n(A)$.

$n(A)$, $n(H) = 13$

if $A = \{x|x$ is a letter in the English alphabet$\}$, then $n(A)=26$.

If $A = \varnothing$ then $n(A) = 0$.



$n(A \cup B) =$ $n(A) + n(B) - n(A \cap B)$

Example

A store has 150 clocks in stock. 100 of these clocks have AM or FM radios. 70 clocks had FM circuitry and 90 had AM circuitry.

a) How many had both AM and FM? (and) $n(AM \cap FM) = 60$

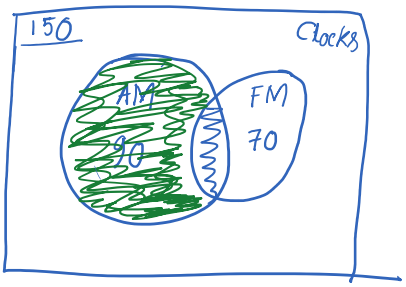
b) How many were AM only? $n(AM \cap FM^c)$

c) How many were FM only?

$= n(AM) - n(AM \cap FM)$

$= 90 - 60 = 30$

$n(FM) - n(AM \cap FM) = 70 - 60 = 10.$



$n(AM \cup FM) = 100$

or

$n(AM) = 90$

$n(FM) = 70$

$n(AM \cup FM) = n(AM) + n(FM) - n(AM \cap FM)$

$100 = 90 + 70 - n(AM \cap FM)$

$n(AM \cap FM) = 90 + 70 - 100$

$= 60$

Example

We are given the following data about the contents of some delivery trucks,

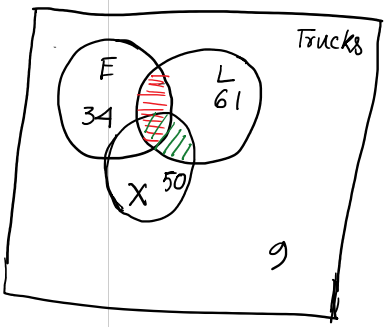
- 34 trucks carried early peaches
- 61 trucks carried late peaches
- 50 trucks carried extra late peaches
- 25 trucks carried early and late peaches
- 30 trucks carried late and extra late peaches
- 8 trucks carried early and extra late peaches
- 6 trucks carried all three kinds of peaches
- 9 trucks carried no peaches

Display this information in a Venn diagram.

How many carried only late peaches?

How many carried only one kind of peaches?

How many trucks went out?



$25 = n(E \cap L)$

$30 = n(X \cap L)$

$8 = n(E \cap X)$

$6 = n(E \cap X \cap L)$

$n(L \cap E^c \cap X^c) = n(L) - n(E \cap L) - n(X \cap L) + n(L \cap X \cap E)$

$$n(L \cup E \cup X) = n(L) - n(E \cap L) - n(X \cap L) + n(L \cap X \cap E)$$

$$= 61 - 25 - 30 + 6 = 12$$

Example

One hundred shoppers are interviewed about the contents of their bags and the following results are found:

- 19 bought Twinkies
- 37 bought diet soda
- 18 bought broccoli
- 1 bought broccoli, diet soda and Twinkies
- 11 bought Twinkies and diet soda
- 0 bought only Twinkies and broccoli
- 24 bought only diet soda

Display this information in a Venn diagram.

Example

Fifty-two people at a home show were surveyed to see if they planned on replacing their kitchen countertops (C), their kitchen floor (F) or their kitchen appliances (A). The following results were found:

- 22 people planned to replace exactly one of these features
- 18 people were planning to replace their kitchen appliances or floor, but not their countertops.
- 6 people planned to only replace their kitchen countertops
- 2 people planned to remodel all three features.
- 25 people planned to replace their countertops.
- 36 people planned to replace their appliances or countertops
- 9 people planned to replace their countertops and floor.

Display this information in a Venn diagram.